

Research on C-V2X synchronous positioning based on weak GNSS signal scenario

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Declaration

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I hereby acknowledge and accept that this thesis may be distributed to future final year students, as an example of the standard expected of final year projects.

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Abstract

In complex urban environments, due to the towering buildings, diverse types of obstacles, and the susceptibility of satellite positioning signals to interference from other wireless communication systems, GNSS rejection and the unavailability of satellite navigation widely exist in application scenarios. When vehicles are driving in weak GNSS scenarios, the positioning accuracy is low, and dynamic positioning cannot even be achieved. With the evolution of the Internet of Vehicles, smart transportation has developed a new mode of vehicle road collaboration, C-V2X, which utilizes wireless signals of specific frequencies to achieve synchronous positioning between vehicles and roadside unit RSU clusters, providing low latency and high-precision vehicle absolute positioning services, and compensating for the shortcomings of traditional satellite navigation in tunnels. Based on MATLAB simulation, this article proposes a method that uses a pseudo satellite RSU unit, combined with inertial measurement instruments and laser ranging to assist in vehicle positioning. It achieves centimetre-sized error positioning and studies the error convergence curve and system positioning performance of vehicles entering tunnels under different motion modes. This article also proposes improvement suggestions for using the GRU model to assist in localization, to address challenges and optimize system performance, guiding practical applications and future research.

Keywords: GNSS, C-V2X, WLS, INS, Kalman filter

Content

1	INTRODUCTION	1-5
1.1	SYNCHRONOUS POSITIONING.....	1-5
1.1.1	Significance of synchronous positioning and research in C-V2X	1-5
1.1.2	Conventional methods of synchronous positioning	1-5
1.1.3	Weak GNSS signal scenario	1-5
1.1.4	C-V2X: Cellular-Vehicle to Everything	1-6
1.2	PROPOSED SYNCHRONOUS POSITIONING SYSTEM FOR WEAK GNSS SIGNAL SCENARIO	1-6
1.2.1	Localization Algorithm Based on RSU-INS Data Fusion and Kalman Filter	1-6
1.2.2	Simulation experiments and performance analysis	1-7
1.2.3	Innovations.....	1-7
2	TECHNICAL BACKGROUND	2-7
2.1	GLOBAL NAVIGATION SATELLITE SYSTEM(GNSS)	2-7
2.2	PSEUDO-SATELLITE	2-8
2.3	ROADSIDE UNIT (RSU)	2-9
2.4	INERTIAL NAVIGATION SYSTEM (INS).....	2-9
2.5	CARRIER PHASE MEASUREMENT.....	2-10
2.6	KALMAN FILTER(KF)	2-10
2.7	GATED RECURRENT UNIT (GRU)	2-11
2.8	GEOMETRIC DILUTION PRECISION (GDOP)	2-12
3	PROBLEMS	3-12
3.1	4-RSU POSITIONING ALGORITHM AND SOLVING OBSERVATION EQUATIONS	3-12
3.2	MASTER STATION SELECTION ALGORITHM	3-12
3.3	DATA FUSION AND FILTER FOR POSITIONING ACCURACY	3-13
4	SOLUTION.....	4-13
4.1	OBSERVATION EQUATION	4-13
4.2	WEIGHTED LEAST SQUARES REGRESSION(WLS).....	4-15
4.3	SIMULATION EXPERIMENT SETTINGS.....	4-15
4.3.1	Noise settings	4-15
4.3.2	Positioning target motion settings.....	4-16
4.3.3	RSU distribution settings	4-16
4.4	MASTER RSU SELECTION ALGORITHM.....	4-17
4.5	POSITIONING SYSTEM	4-18
4.5.1	System with DEM and Laser range (RSU-LR-DEM)	4-18
4.5.2	System with INS and Kalman Filter (INS-KF).....	4-18
5	EVALUATION.....	5-19
5.1	EVALUATING INDICATOR GDOP	5-19
5.2	POSITION AND VELOCITY ESTIMATION.....	5-20
5.3	ESTIMATION ERROR AND TIME CONSUMPTION	5-21
6	CONCLUSION.....	6-22
6.1	SYSTEM PERFORMANCE EVALUATION	6-22
6.2	FURTHER PROBLEM	6-22
6.3	FUTURE IMPROVEMENT	6-23
7	REFERENCES	7-24
8	APPENDICES	8-25

1 Introduction

1.1 Synchronous positioning

1.1.1 Significance of synchronous positioning and research in C-V2X

Synchronous positioning refers to the simultaneous determination of the position of objects in space-time, often using signals from multiple satellites or a network of fixed stations. Its significance in modern technology and society is vast, as precise positioning is fundamental to a multitude of applications[1]. In the context of intelligent transportation systems and vehicle positioning, synchronous positioning is a cornerstone that enables enhanced safety, efficiency, and automation. It allows vehicles to determine their location accurately in relation to each other and to the infrastructure, which is critical for navigation, traffic management, and the development of autonomous driving technologies.

The evolution of intelligent transportation leverages this precise positioning to optimize traffic flow, reduce congestion, and improve overall transportation safety. It also plays a significant role in enabling vehicles to communicate with each other and with roadside infrastructure, a concept known as Cooperative Vehicle-to-Everything (C-V2X) communication[2]. The research in C-V2X is significant because it is poised to revolutionize how vehicles interact within the transportation ecosystem[3]. By ensuring reliable and real-time positioning, vehicles can share their location, speed, and direction with nearby vehicles and infrastructure, facilitating a collaborative environment where each participant is aware of the others. This leads to an increase in situational awareness and can significantly reduce accidents, improve traffic efficiency, and pave the way for the next generation of automated driving systems.

In summary, the significance of synchronous positioning in the realm of smart transportation and C-V2X research lies in its ability to provide a robust foundation for safety and efficiency improvements in transportation systems, contributing to the advancement of autonomous driving and the realization of smarter, more connected cities.

1.1.2 Conventional methods of synchronous positioning

Conventional methods of synchronous positioning include several key techniques, each designed for specific positioning and navigation requirements. Classify based on the number of satellites in use, Conventional methods include Single-Satellite Positioning and Multi-Satellite Positioning. Single-Satellite Positioning relies on a single satellite to provide a position fix by measuring the time delay or carrier phase of a known signal[4]. It's less common due to its limited accuracy and reliance on external information to resolve position ambiguities. Multi-Satellite Positioning utilizes multiple satellite signals to determine position by intersecting the spherical surfaces of signal propagation. It typically requires a minimum of 4 satellites to resolve a position in three dimensions and time, making it the foundational method for systems like GPS, GLONASS, Galileo and Beidou.

Measurement methods in positioning systems are typically categorized into two types: Pseudo-range Measurement and Carrier Phase Measurement. The former calculates distances from satellites using the time lag of signal reception, adjusting for discrepancies between the clocks on the satellites and the receiver to pinpoint its location. These calculations hinge on the precise timing of signal dispatch and arrival, utilizing the known velocity of signal propagation (the speed of light). The primary timing methodologies encompass Time of Arrival (TOA), Time Difference of Arrival (TDOA), and Time-Division Multiple Access (TDMA). Meanwhile, Carrier Phase Measurement relies on the detailed phase information of the carrier wave for more refined distance assessments from satellites to the receiver. This method can yield higher accuracy in positioning but necessitates intricate processing to clarify the initial phase count's integer ambiguity.

1.1.3 Weak GNSS signal scenario

Weak GNSS signal scenarios occur when the power of the satellite signals that reach the receiver is low, often too low for traditional receivers to process effectively[5]. These scenarios can be caused by a number of factors, such as: Obstructions (Buildings, mountains, or heavy foliage can block or deflect the line-of-sight path from the satellite to the receiver); Atmospheric Conditions (Ionospheric and tropospheric disturbances can weaken the signals as they travel from space to Earth); Indoor Environments; Urban Canyons.

In weak signal conditions, traditional positioning methods face several challenges, such as: Receivers can lose the lock on the signal, leading to interruptions in tracking and increased time to first fix; Weak signals are more susceptible to noise, which can degrade the accuracy of the calculated position. In environments with obstructions, reflected signals can be as strong as or stronger than the direct signal, leading to errors that are difficult to correct; Conventional receivers may not be sensitive enough to detect and process weak signals, leading to failure to determine a position fix. In this case, Advanced techniques and the integration of GNSS with other technologies are often used to mitigate these issues and provide reliable positioning even in scenarios where GNSS signals are weak.

1.1.4 C-V2X: Cellular-Vehicle to Everything

Cellular Vehicle-to-Everything communication(C-V2X), refers to a wireless communication technology that allows vehicles to directly collaborate with other vehicles ,pedestrians, infrastructure, and the network, forming a comprehensive ecosystem known as V2X. This communication is facilitated by cellular technology, including the existing LTE networks and the forthcoming 5G networks, as shown in Fig 1-1.



Fig 1-1 C-V2X Schematic diagram

The primary goals of C-V2X are to enhance road safety, improve traffic efficiency, and pave the way for autonomous vehicles[3]. By allowing direct, low-latency communication between the different elements of the traffic system, C-V2X helps in several ways. C-V2X enables vehicles to share information about their speed, direction, and braking status, which can warn drivers about potential collisions or hazards ahead, even beyond the line of sight or in poor visibility conditions; By communicating with traffic signals and sensing traffic patterns, C-V2X can help optimize driving routes to reduce congestion and enhance fuel efficiency; For self-driving cars, C-V2X provides critical real-time data from the surrounding environment, necessary for decision-making processes.

C-V2X is considered a key enabler of smart transportation systems of the future, as it works to create a network where vehicles and infrastructure work together seamlessly.

1.2 Proposed synchronous positioning system for weak GNSS signal scenario

1.2.1 Localization Algorithm Based on RSU-INS Data Fusion and Kalman Filter

This article proposes an RSU-INS real-time positioning system based on C-V2X mode for weak GNSS scenarios in tunnels. The system uses RSU and specific frequency bands, DEM, edge measuring

laser range, and IMU as sensors, referring to the 4-star positioning principle of GNSS, to locate the target vehicle traveling at high speed in three-dimensional space. The main body of the system uses data from multiple sensors, and uses weighted least squares(WLS) regression algorithm for data fusion and total differential linear equation solving. On the premise of pre arranging the RSU cluster, the system also includes a dynamic selection algorithm for the master station. The system also uses Kalman filtering to filter the solution data by setting process noise and observation noise, thereby improving positioning accuracy and credibility. Finally, the dynamic positioning system proposed in this article can achieve centimetre-level positioning accuracy, meeting the lane-level positioning accuracy requirements in the fields of C-V2 and autonomous driving. In addition, this article also preliminarily discusses and practices the feasibility of using Gated Recurrent Unit (GRU) machine learning methods for pseudo satellite and C-V2X positioning.

1.2.2 Simulation experiments and performance analysis

This article conducts simulation experiments in the MATLAB environment, setting up straight tunnels and curved tunnels with a length of 2km, a width of 20m, and a height of 10m, to locate the target vehicle through the tunnel in a uniform and accelerated motion. GDOP and computational time consumption are used as evaluation indicators to analyse the positioning performance of the positioning system. In the simulation experiment, a set error function similar to Gaussian distribution is used as the noise simulation sensor error. This article conducts simulation experiments on RSU-DEM-Laser range, RSU-INS, and positioning systems with Kalman filtering, and analyses the positioning performance of the system by comparing the GDOP error curve. Finally, the article also proposes tunnel positioning scenarios and evaluates the feasibility of using GRU for C-V2X positioning scheme by analysing the training loss function and mean square error of the GRU model.

1.2.3 Innovations

- a) **Weak GNSS signal scenarios positioning:** Proposed vehicle positioning solutions in weak GNSS or even GNSS rejection environments, achieving high-precision and low latency multi data fusion positioning systems.
- b) **Master station selection algorithm:** A feasible arrangement and distribution scheme for RSU clusters is proposed for straight tunnels and curved (a segment of arc) tunnels, and a dynamic selection algorithm for main master stations is proposed, achieving a transitional connection between GNSS positioning and pseudo satellite positioning.
- c) **Data fusion:** Multiple traditional positioning sensors and devices were used for data fusion, combined with Kalman filtering for denoising, significantly improving positioning accuracy and credibility.
- d) **Machine leaning:** Proposed a proposal for applying the GRU model to C-V2X, combining machine learning with vehicle localization.

2 Technical Background

2.1 Global Navigation Satellite System(GNSS)

Global Navigation Satellite System(GNSS) includes global, regional, and enhanced. GNSS is a global autonomous geospatial positioning satellite system used for navigation and positioning measurement. GNSS system uses satellite signals to transmit real-time position and time information, and thus calculates geographic location information such as longitude and latitude of ground receiving equipment[7]. The world's first artificial satellite was Sputnik, developed by the Soviet Union. Three years later, the US military developed the world's first Doppler effect based positioning system, Transit. Afterwards, in order to improve positioning accuracy, the United States developed a more accurate Timation based on satellite

position and specific time precise ranging, which is also considered a pioneer of Global Positioning System(GPS). In 1973, the United States established a GPS infrastructure and gradually improved it; In 1982, the GLONASS system was finally formed after more than ten years of research and development; In 2000, China introduced the Beidou system; In 2010, Japan released the Quasi Zenith Satellite System (QZSS); In 2005, the European Union launched the Galileo system; In 2013, India launched the Indian Regional Navigation Satellite System (IRNSS), now known as NavIC.

GNSS positioning is based on the principle of triangulation and relies on the estimation of the distance between the receiver and each visible satellite, namely the three-point positioning method. The intersection point of three circles in space is the positioning position (in fact, in mathematical problems, there may be more than one intersection point of three circles, but the rest will be considered abnormal, such as not on the Earth's surface). From this perspective, positioning requires at least three satellites. However, there are differences in practical applications: because GNSS signals need to propagate a very long distance, there is a lot of interference and attenuation during this period, so the unified time reference error is very large. Therefore, in practical applications, time is used as the fourth variable: the time difference between the receiver reference and the satellite onboard clock.

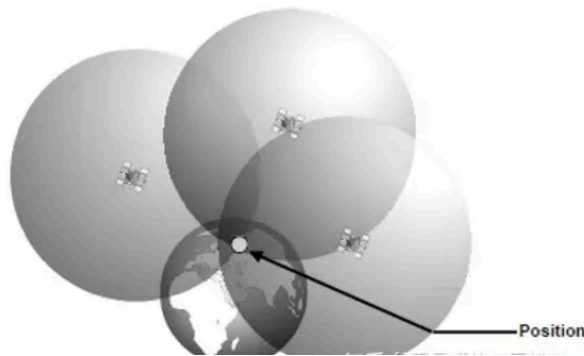


Fig 2-1 three-point positioning method

2.2 Pseudo-satellite

Taking the GPS satellite navigation system as an example, the GPS pseudo satellite positioning system is a regional positioning system that simulates the GPS positioning system. The basic theory and research methods are derived from GPS, and more than 4 GPS pseudo satellites can be used as signal sources to simulate the satellites in the GPS system. Independent coordinate systems and time standards are used to form a pseudo satellite network, and the synchronization control of the pseudo satellite system is completed through the GPS receiver and the main control station. The reception of navigation signals and the extraction and calculation of positioning information can be carried out in a certain area. The pseudo satellite observation equation is different from the conventional GPS observation equation[8]: firstly, since pseudo satellites are generally placed within the atmospheric troposphere, there is no need for ionospheric delay correction; Secondly, the path propagated by pseudo-satellite signals generally propagates at lower altitudes in the troposphere, so the commonly used tropospheric correction models for GPS observations cannot be directly used. Instead, special tropospheric models must be used for correction, or the observations of the pseudo-satellite itself can be used for correction. Finally, due to the fact that the clock error of the pseudo-satellite is much larger than that of the GPS satellite, correction must be made during measurement. The accuracy, reliability, and integrity of GPS positioning technology are closely related to the number and geometric shape of GPS satellites being tracked. However, in some special locations, such as deep mountain canyons, mine tunnels, or large urban buildings, due to severe signal obstruction, there is often no legal position or positioning accuracy that does not meet user requirements, which will reduce

the availability of the GPS system. In this case, using pseudo satellite positioning technology can better meet user needs.

2.3 Roadside Unit (RSU)

The Roadside Unit (RSU) is a significant component of the vehicle road collaboration roadside end. It is applied to collect current road conditions and traffic conditions, communicate with roadside perception devices, traffic lights, electronic signs and other terminals through communication networks, achieve vehicle road interconnection and real-time interaction of traffic signals, assist drivers in driving, and ensure the safety of personnel and vehicles in the entire transportation field, as shown in Fig 2-2 and Fig 2-3.

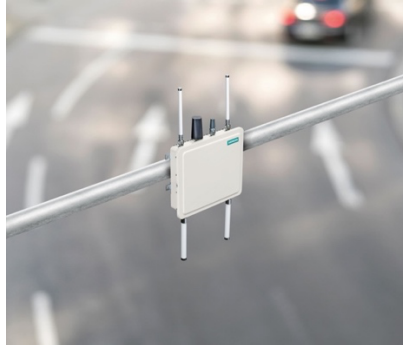


Fig 2-2 Roadside Unit



Fig 2-3 vehicle road collaboration

2.4 Inertial Navigation System (INS)

Inertial Measurement Unit (IMU) is a device used to measure the acceleration and angular velocity of an object. IMU typically consists of 3 accelerometers and 3 gyroscopes, used to measure the acceleration and angular velocity of an object on 3 axes, as shown in Fig 2-4. IMU is commonly used for measuring and controlling equipment such as aircraft, navigation systems, robots, etc. Inertial Navigation System (INS) uses IMU measurement data for navigation. INS uses the acceleration and angular velocity information measured by IMU to estimate the position, velocity, and direction of objects. INS is commonly used in applications that require high-precision navigation, such as aviation and missiles.

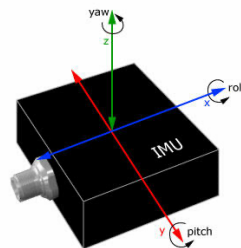


Fig 2-4 IMU device

INS is a fully autonomous navigation system has high short-term measurement accuracy. In addition to providing position and speed, it can also provide attitude information. However, due to the internal integration of the algorithm, the error of the inertial sensor will continue to accumulate, resulting in unlimited growth of long-term navigation errors. Integrating INS with GPS leverages the complementary strengths of both, resulting in a navigation solution that surpasses the stability and precision achievable by either system alone. The INS/GPS integrated navigation system can output high-frequency navigation parameter information (position, speed, attitude), and can achieve high accuracy in both long and short term navigation processes[6]. By using the optimal estimation method based on Kalman filtering to fuse GPS and INS positioning navigation information, reliable navigation solutions can be obtained. GPS can

prevent inertial data drift, and INS can provide position, velocity, and attitude information in the event of GPS signal interruption. A typical INS/GPS combination is shown in Fig 2-5:

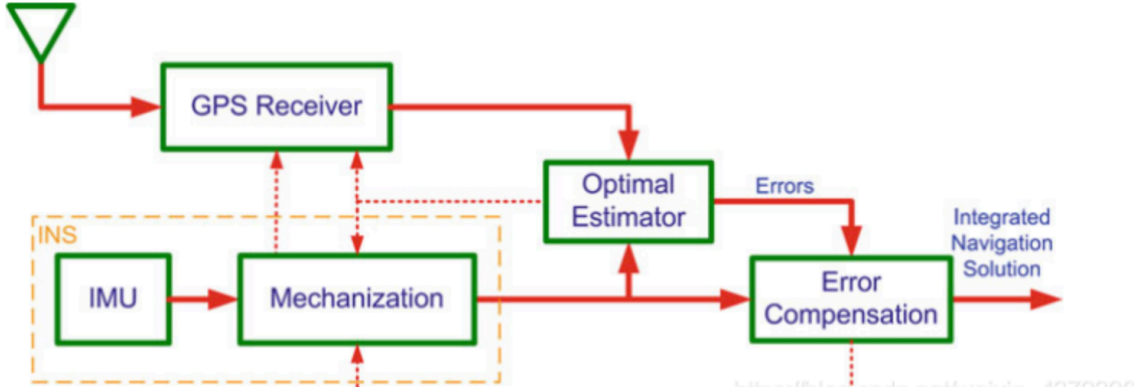


Fig 2-5 typical INS/GPS combination navigation system

2.5 Carrier phase measurement

Carrier phase measurement is a technique used in GNSS or pseudo satellite positioning system to determine the precise distance between a transmitter and a receiver[9]. This method is considerably more accurate than the basic pseudo-range measurements, which are based on the timing of signal travel. Carrier phase measurement signals are composed of two main components: the carrier wave, which is a high-frequency signal, and the modulation, which usually contains the navigation message and codes. The method determines how many wavelengths signal has traveled by measuring the phase changes in the carrier wave[10]. One challenge with carrier phase measurement is resolving the initial phase ambiguity – that is, determining the exact number of full wavelengths between the satellite and the receiver, which is not known from the phase measurement alone.

2.6 Kalman filter(KF)

Kalman filtering is a recursive algorithm based on the minimum mean square error criterion. It establishes a mathematical model based on the state of signals and noise, and can perform stable linear unbiased estimation of system states containing random errors[11]. The algorithm essentially reduces the impact of errors and random interference through the "estimation measurement update" step. Each step of the recursive algorithm is mainly divided into two stages: the prior estimation stage and the update stage (posterior estimation). The prior estimation stage is to obtain the current prior estimation value through the state equation, provided that the mathematical model is accurate; The update stage involves further integrating state observations, correcting prior estimates based on Kalman gain, and obtaining updated state estimates[12].

Assuming X_k is the true value of the state, $\hat{X}_{k|k}$ is the k -th state estimation value of the Kalman filtering algorithm, using $X_k - \hat{X}_{k|k}$ represents the posterior state error estimation, and the algorithm objective is to minimize $E \left[\|X_k - \hat{X}_{k|k}\|^2 \right]$, which is equivalent to minimizing the trace of the posterior estimation covariance matrix $P_{k|k}$, can be derived from which the optimal Kalman gain for the k -th step is:

$$K_k = \frac{P_{k|k-1} H_k^T}{H_k P_{k|k-1} H_k^T + R_k} \quad (2.6.1)$$

In the formula: $P_{k|k-1}$ is the covariance matrix prior of the $k - 1$ -th estimation stage. Estimate; H_k is the observation matrix; R_k is the covariance matrix of the observed noise. The optimal Kalman filtering algorithm is specific to following equations

Prior estimation stage:

$$\hat{X}_{k|k-1} = F_k \hat{X}_{k-1|k-1} + B_k u_k + w_k \quad (2.6.2)$$

$$P_{k|k-1} = F_k P_{k-1|k-1} F_k^T + Q_k \quad (2.6.3)$$

In the formula: $\hat{X}_{k|k-1}$ is the k -th step prior estimate value; $\hat{X}_{k-1|k-1}$ is the optimal estimation of the state in the $k - 1$ -th update stage, $P_{k|k-1}$ is the prior estimate of the covariance matrix for the $k - 1$ -th step, $P_{k-1|k-1}$ is the optimal covariance estimate for the $k - 1$ -th update stage, F_k is the state matrix; u_k is the control quantity, B_k is the control matrix of u_k , $w_k \sim N(0, Q_k)$ is the process noise; Q_k is the covariance matrix of process noise.

Update stage:

$$\hat{X}_{k|k} = \hat{X}_{k|k-1} + K_k [z_k - H_k \hat{X}_{k|k-1}] \quad (2.6.4)$$

$$P_{k|k} = [I - K_k H_k] P_{k|k-1} \quad (2.6.5)$$

In the formula, $\hat{X}_{k|k}$ is the optimal estimated value of the state in the k -th update stage. z_k is the measurement value for the k -th state, K_k is the optimal Kalman gain for k -th step.

Through iterative formulas, it can be seen that Kalman filtering, as a recursive calculation method, is relatively simple in calculation. During the iteration process, only the calculation parameters of the previous step need to be stored, occupying less storage space.

2.7 Gated Recurrent Unit (GRU)

A Gated Recurrent Unit (GRU) is particularly well-suited for processing sequential data, offering improvements over traditional recurrent neural networks (RNNs) by addressing the vanishing gradient problem through its gating mechanisms[13]. GRU utilizes a pair of gates to manage the internal information flow. The update gate is responsible for the extent to which previous information is retained for future use, and the reset gate is in charge of determining the quantity of previous information that should be disregarded. These gates are adaptive and learn how to optimally maintain and discard information through backpropagation, allowing GRUs to capture dependencies for different lengths of time within a sequence. This makes GRUs advantageous in tasks like time series prediction, natural language processing, and speech recognition, where the ability to capture long-term dependencies is crucial for performance[14]. Unlike their predecessor LSTM (Long Short-Term Memory), GRUs are simpler in terms of computational architecture but still offer comparable performance, often being the architecture of choice due to their efficiency in training and inference.

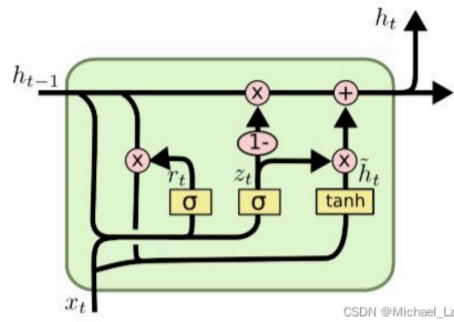


Fig 2-6 simplified GRU structure

The simplified GRU structure is shown in the Fig 2-6. h_{t-1} is the hidden state of last state, containing information about data previously seen by nodes. h_t is the hidden state passed to the next moment, $\tilde{h}_t$ is candidate hidden state, x_t is input information at the current moment. r_t is Reset Gate and z_t is Update Gate. σ is the sigmoid function, through which data can be transformed into numerical values in the range of $[0, 1]$. The tanh function, through which data can be transformed into numerical values within the range of $[-1, 1]$.

2.8 Geometric Dilution Precision (GDOP)

The Geometric Dilution Precision (GDOP) is a key indicator for positioning accuracy, representing the distance vector amplification factor between the receiver and space satellites caused by GPS ranging errors. The volume associated with the unit vector extending from the receiver to the orbiting satellite, indicative of the positioning solution, is proportionately related to the Geometric Dilution of Precision (GDOP)[15]. This volume, inversely correlated with GDOP, serves as a measure of geometric accuracy. Essentially, a higher GDOP value corresponds to a reduced volume of this vector space, due to the narrow angular separation between the receiver and satellites. Consequently, a rise in GDOP values is synonymous with diminished locational precision. An optimal GDOP, characterized by a lower numerical value, denotes a broader volume of the unit vector, enhancing the accuracy of positioning. The ideal geometric condition is achieved when the satellites are not clustered in a single region but are dispersed across various azimuth sectors in space, facilitating higher positioning accuracy.

Situations with poor and good geometric accuracy factors for positioning are shown in Fig 2-6 and Fig 2-7. Usually, the ranging error of satellite signals multiplied by an appropriate GDOP value can estimate the obtained position or time error. Different GDOP values are calculated from the covariance matrix of navigation.

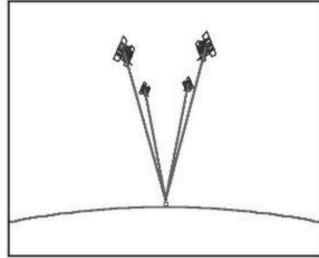


Fig 2-6 poor geometric accuracy factors

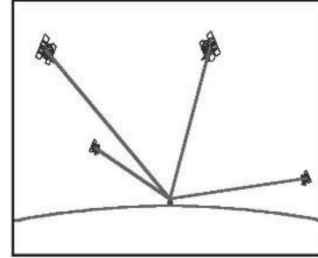


Fig 2-7 good geometric accuracy factors

3 Problems

3.1 4-RSU positioning algorithm and Solving observation equations

The commonly used method for GNSS positioning is four satellite carrier measurement positioning, which simultaneously uses four satellites to send signals to the target and measure the carrier phase difference when receiving the signal, and settles the positioning coordinates by establishing observation equations. In pseudo satellite systems, similar positioning methods can still be used, with the difference being the use of low orbit satellites or sensors deployed in the environment. In C-V2X mode, we can use RSUs as sensors (or observation devices) and observe using four RSUs at the same time. However, how to solve the observation equation remains a problem that we need to face. In addition, in the process of data fusion, how to solve more complex observation equations will also be discussed in this article.

3.2 Master Station Selection Algorithm

In environments with weak GNSS or GNSS rejection, RSU clusters are composed of a large number of RSU devices distributed around the environment. For example, in tunnels, vehicles move along the central axis of the road, and RSUs are distributed on the top part of the tunnel and along the axis, as shown in Fig 3-1. However, during the process of vehicle movement, RSU clusters composed of specific RSU devices locate the signals sent by the vehicle at the same time, rather than all RSU clusters sending signals. At this time, the RSU that centrally processes the RSU clusters becomes the master station. The selection of the main master station can serve as a substation for data processing, and is also crucial in unifying the clock of RSU clusters. Once the main master station is determined, the location and distance information

of the RSU cluster can be obtained for localization. How to choose the main master station and the dynamic selection algorithm of the main master station is crucial for the positioning performance, stability, and computational efficiency of the positioning algorithm. Due to the similarity in analysis methods between the vertical and horizontal master station selection problems in three-dimensional space, this article focuses on the dynamic selection algorithm of the main master station in the horizontal direction to simplify the problem.

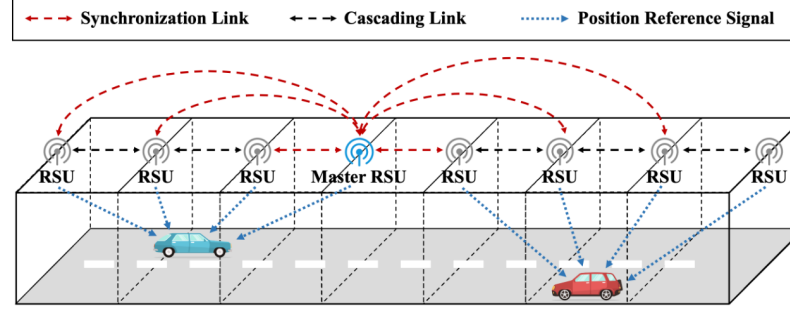


Fig 3-1 Master RSU Synchronization

3.3 Data fusion and filter for positioning accuracy

Multiple sensors and positioning methods are used in the system, and data fusion and filtering are crucial. Vehicle positioning requires a positioning accuracy of lane level, with an error of centimetre-level, otherwise it cannot meet the safety requirements of vehicle driving and navigation. How to perform data fusion to more accurately calculate the coordinates of the target and improve the efficiency and stability of the system is the issue that needs to be discussed in this article. In addition, it is also important to set the measurement noise and process noise of the Kalman filter to reduce the error and make the filtered results more reliable.

In positioning algorithms, the least squares regression algorithm is the most fundamental and important part, and it is necessary to go through the least squares regression algorithm in each iteration. Therefore, the computational efficiency, stability, and accuracy of the least squares regression algorithm are crucial. Low computational efficiency not only increases the latency of positioning, but also causes excessive load on the memory space of onboard equipment; Stability is related to positioning accuracy and the sustainability of the positioning system; Accuracy affects the positioning accuracy and base station selection algorithm. If high-precision calculations cannot be achieved, the regression algorithm is prone to non-convergence. To solve the above problems, the optimization of the least squares regression algorithm and the selection of initial values for calculation are crucial.

4 Solution

4.1 Observation Equation

Similar to the pseudo-range in GNSS, the pseudo-range between the target and a certain RSU device has the following expression[16]:

$$\tilde{p} = \sqrt{(X^s - X)^2 + (Y^s - Y)^2 + (Z^s - Z)^2} + Cdt - CdT + \delta_{atmos} \quad (4.1.1)$$

where $\tilde{p}$ is Pseudo-range observations, (X^s, Y^s, Z^s) is the coordinates of a certain RSU device, (X, Y, Z) is the coordinates of target, C represents the speed of light, dt is receiver (target) clock error, dT is RSU clock deviation, δ_{atmos} is the error caused by the atmosphere. But in C-V2X, by RSU cascading and error caused by atmosphere is very tiny, so dT and δ_{atmos} can be ignorable.

$$\tilde{p} = \sqrt{(X^s - X)^2 + (Y^s - Y)^2 + (Z^s - Z)^2} + Cdt \quad (4.1.2)$$

Linearize the nonlinear equation using Taylor's formula:

$$\tilde{p} \approx \sqrt{(X^s - X)^2 + (Y^s - Y)^2 + (Z^s - Z)^2} + Cdt_0 + \frac{\partial \tilde{p}}{\partial X}|_{x_0} \Delta X + \frac{\partial \tilde{p}}{\partial Y}|_{y_0} \Delta Y + \frac{\partial \tilde{p}}{\partial Z}|_{z_0} \Delta Z + \frac{\partial \tilde{p}}{\partial (Cdt)} \Delta(Cdt) \quad (4.1.3)$$

For a specific RSU, we determine that:

$$R_0 = \sqrt{(X^s - X)^2 + (Y^s - Y)^2 + (Z^s - Z)^2} \quad (4.1.4)$$

$$a_x = \frac{\partial \tilde{p}}{\partial X}|_{x_0} = -\frac{X^s - X_0}{R_0} \quad (4.1.5)$$

$$a_y = \frac{\partial \tilde{p}}{\partial Y}|_{y_0} = -\frac{Y^s - Y_0}{R_0} \quad (4.1.6)$$

$$a_z = \frac{\partial \tilde{p}}{\partial Z}|_{z_0} = -\frac{Z^s - Z_0}{R_0} \quad (4.1.7)$$

For the i -th RSU, there are expressions for pseudo-range and residual as follows:

$$\tilde{p}_i \approx R_{0i} + Cdt_0 + a_{xi} \Delta X + a_{yi} \Delta Y + a_{zi} \Delta Z \quad (4.1.8)$$

$$v_i = R_{0i} + Cdt_0 + a_{xi} \Delta X + a_{yi} \Delta Y + a_{zi} \Delta Z - \tilde{p}_i \quad (4.1.9)$$

$$\tilde{p}_i = (\varphi + N)\lambda \quad (4.1.10)$$

Where N is the Ambiguity of whole cycles, λ is the wavelength of the frequency band. As shown above, by measuring the carrier phase and processing the integer ambiguity, we can obtain the relationship between pseudo-range, $(\Delta X, \Delta Y, \Delta Z)$ and residual.

After employing ambiguity resolution method, use precise carrier phase measurements for following models. The linearized observation equation is:

$$L_{vps} + V_{vps} = A_{vps} \Delta X \quad (4.1.11)$$

where, L_{vps} denotes carrier phase measurements, V_{vps} is the residual vector, X is the unknown vector and A_{vps} is the coefficient matrix. Additionally:

$$L_{vps} = \begin{bmatrix} R_{01} + Cdt_0 - \tilde{p}_1 \\ R_{02} + Cdt_0 - \tilde{p}_2 \\ \vdots \\ R_{0n} + Cdt_0 - \tilde{p}_n \end{bmatrix}, V_{vps} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}, A_{vps} = \begin{bmatrix} a_{x1} & a_{y1} & a_{z1} & 1 \\ a_{x2} & a_{y2} & a_{z2} & 1 \\ \vdots & \vdots & \vdots & \vdots \\ a_{xn} & a_{yn} & a_{zn} & 1 \end{bmatrix}, \Delta X = \begin{bmatrix} \Delta X \\ \Delta Y \\ \Delta Z \\ \Delta(Cdt) \end{bmatrix} \quad (4.1.12)$$

The laser ranging(LR) module measures the range between the vehicle and the tunnel sidewall. The lateral position can be calculated by the two-side range measurements. As shown in Fig 4-1, based on the homothetic triangle theory, the formula of the lateral position y_{pr} is as follows:

$$y_{pr} = \frac{d_1 w}{d_1 + d_2} \quad (4.1.13)$$

where, w is the width of tunnel, d_1 and d_2 are the measurements of LR;

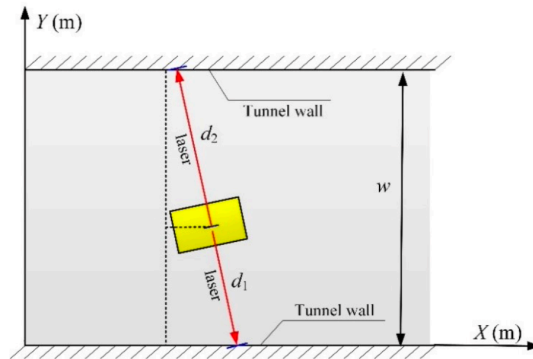


Fig 4-1 Schematic diagram of lateral positioning estimation

The quasi observation L_{pr} corresponding to the priori information can be written as:

$$L_{pr} + V_{pr} = A_{pr} \Delta X \quad (4.1.14)$$

where, A_{pr} and V_{pr} are the design matrix and residual vector of L_{pr} , and:

$$A_{pr} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \quad (4.1.15)$$

$$L_{pr} = [y_{pr} \quad z_{pr}]^T \quad (4.1.16)$$

where, y_{pr} and z_{pr} is the priori lateral position derived by LR and the priori height derived by DEM, respectively.

Thus, the total observation equation is:

$$\begin{bmatrix} V_{vps} \\ V_{pr} \end{bmatrix} = \begin{bmatrix} L_{vps} \\ L_{pr} \end{bmatrix} - \begin{bmatrix} A_{vps} \\ A_{pr} \end{bmatrix} \Delta X \quad (4.1.17)$$

4.2 Weighted Least Squares regression(WLS)

Employ weighted least squares (WLS) to estimate the optimal solution ΔX of observation equation:

$$\Delta X = (A_{vps}^T P_{vps} A_{vps})^{-1} (A_{vps}^T P_{vps} L_{vps}) \quad (4.2.1)$$

or

$$\Delta X = (A_{vps}^T P_{vps} A_{vps} + A_{pr}^T P_{pr} A_{pr})^{-1} (A_{vps}^T P_{vps} L_{vps} + A_{pr}^T P_{pr} L_{pr}) \quad (4.2.2)$$

And the covariance matrix of the WLS estimator can be derived as:

$$Q_x = (A_{vps}^T P_{vps} A_{vps}) \quad (4.2.3)$$

or

$$Q_x = (A_{vps}^T P_{vps} A_{vps} + A_{pr}^T P_{pr} A_{pr}) \quad (4.2.4)$$

where, Q_x is the covariance matrix of the estimate ΔX . P_{vps} and P_{pr} are the weight matrix of the VPS observation L_{vps} and the prior observation L_{pr} , respectively. In this article, we set weight matrix as:

$$P_{vps} = V_{vps}^{-2}, P_{pr} = V_{pr}^{-2} \quad (4.2.5)$$

Perform weighted least squares regression multiple times to calculate the optimal ΔX solution for each step, thereby continuously approaching the optimal solution X based on the initial values X_{hat} and ΔX , until the solution of X hardly changes(that is ΔX approaches 0), the process is shown in Fig 4-2.

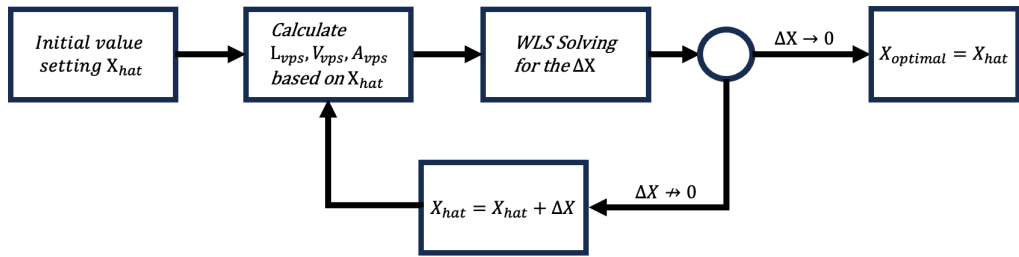


Fig 4-2 WLS iteration for optimal ΔX solution

4.3 Simulation experiment settings

4.3.1 Noise settings

In the simulation experiment, an error function similar to a Gaussian distribution is used as noise[17]. Assuming the error is δ , the maximum error is set to δ_0 . It is considered that the error satisfies a partial normal distribution $N(0, \delta_0^2)$. The error distribution function is set as follows:

$$\delta(\delta_0) = \begin{cases} -\delta_0, & \delta \leq -\delta_0 \\ \delta_0, & \delta \geq \delta_0 \\ \delta \sim N(0, \delta_0^2), & -\delta_0 < \delta < \delta_0 \end{cases} \quad (4.3.1)$$

In the simulation experiment, the RSU carrier phase measurement value, DEM height measurement value, Laser range measurement value, IMU speed and acceleration measurement value are all theoretical values plus simulated real measurement values of noise following this error function. In the simulation experiment of this article, the maximum error of carrier phase measurement value is set to 5m, the maximum error of edge Laser range is 0.1m, and the maximum error of DEM is 0.05m. The error of IMU equipment in measuring acceleration is not considered for the time being.

4.3.2 Positioning target motion settings

Mode_1: The scene is a straight and uniform motion of the target, passing through a straight tunnel. The straight tunnel is 2000m long, 20m wide, and 10m high, with two lanes in opposite directions, each 10m wide. The target initial speed is 60km/h, with the speed direction parallel to the centerline of the lane and no acceleration in any direction. The initial position is in the right lane and 7m away from the centerline of the road. The vehicle height is 1m, and it just enters the tunnel at the initial moment. Assuming that there is still a GNSS signal before the vehicle enters the tunnel, the initial motion information is provided by GNSS, and the subsequent positioning is carried out by the C-V2X system. The positioning process lasts for 120 s, and the vehicle has already exited the tunnel.

Mode_2: Perform linear acceleration motion on the scene as the target, passing through the same straight tunnel as Mode1. The target initial speed is 11m/s, with a speed direction parallel to the centerline of the lane. There is an acceleration of $0.1m/s^2$ in the speed direction and no acceleration in other directions. The initial position is in the right lane and 7m away from the centerline of the road. The vehicle height is 1m, and it just enters the tunnel at the initial moment. Assuming that there is still a GNSS signal before the vehicle enters the tunnel, the initial motion information is provided by GNSS, and the subsequent positioning is carried out by the C-V2X system. The positioning process lasts for 120s, and the vehicle has already exited the tunnel.

Mode_3: Perform curved motion on the scene as the target, passing through a curved tunnel. The circular arc tunnel is 2000m long, 20m wide, and 10m high, with two opposite lanes, each 10m wide. The centerline of the road is a circular arc with a radius of 10km and a center angle of 0.2 rad, with both sides of the tunnel parallel to the centerline. The initial speed of the target is 60km/h, and the direction of the initial speed is parallel to the centerline of the lane. There is a centripetal acceleration pointing towards the center of the motion trajectory, which causes the target to move in a uniform circular motion. The initial position is in the right lane and 7m away from the centerline of the road, with a vehicle height of 1m. At the initial moment, it happens to enter the tunnel. Assuming that there is still a GNSS signal before the vehicle enters the tunnel, the initial motion information is provided by GNSS, and the subsequent positioning is carried out by the C-V2X system. The positioning process lasts for 120 s, and the vehicle has already exited the tunnel.

4.3.3 RSU distribution settings

At the same time, 4 RSUs are used for positioning, using the L1 frequency band with a center frequency of $f_1 = 1575.42 \times 10^6$, wavelength $\lambda = 0.19029367$, and a constant light speed of $C = 3 \times 10^8$. A total of 24 RSUs are used in all modes.

Mode_1 and Mode_2: Due to the significant correlation between the distribution of RSUs and tunnel and target motion trajectories, the RSU arrangement distribution of Mode_1 and Mode_2 in the simulation experiment is consistent. In a straight tunnel with a length of 2000m, the RSU cluster is evenly distributed along the centerline direction of the road, with equipment spaced at 100m intervals and all located at the top of the tunnel. The first RSU is arranged 100m before entering the tunnel, and the last tunnel is arranged 200m after exiting the tunnel, for signal transition of C-V2X and GNNS. The simulation scenario is shown in the Fig 4-3.

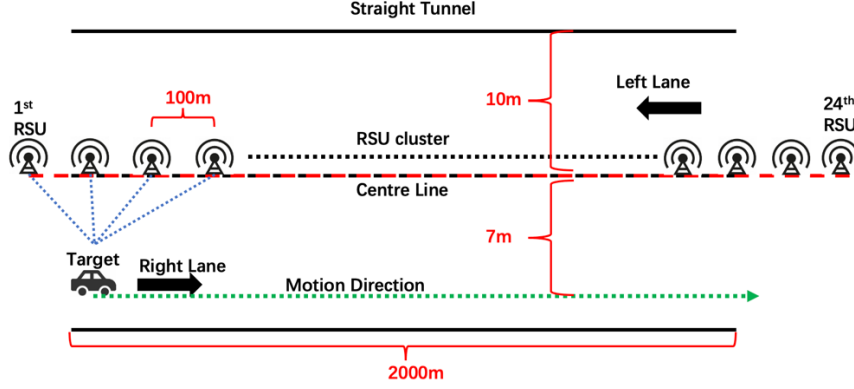


Fig 4-3 Mode_1 and Mode_2 Setting diagram

Mode_3: In a 2000m long circular tunnel, the RSU cluster is evenly distributed along the centreline direction of the road, with a distance of 0.01rad between the centre angles of the equipment and all located at the top of the tunnel. The first RSU is arranged at 0.01 rad before entering the tunnel, and the last tunnel is arranged at 0.02 rad after exiting the tunnel, used for signal transition of C-V2X and GNNS. The simulation scenario is shown in Fig 4-4.

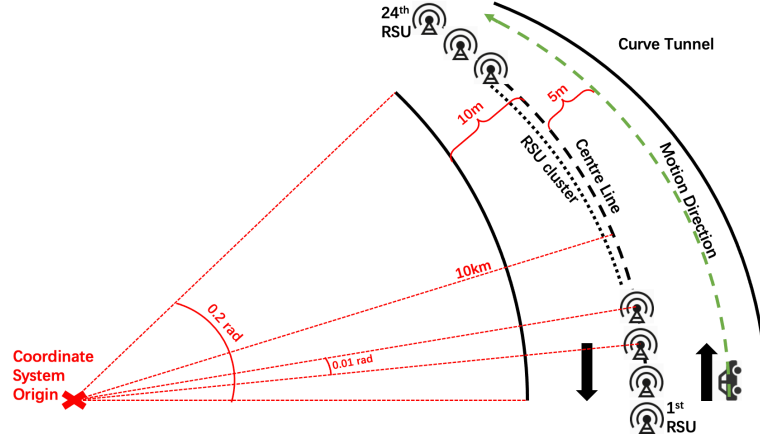


Fig 4-4 Mode_3 Setting diagram

4.4 Master RSU Selection Algorithm

At the same time, use 4 RSUs to locate the target. Before each localization estimation iteration, the previous estimation result is used as the initial estimation state, and the Master RSU Selection is performed based on the relative position relationship between the target estimated initial state and the RSU cluster. If the target and RSU meet the following relationship:

In Mode_1 and Mode_2:

$$x \geq x_i \wedge x < x_{i+1}$$

In Mode_3:

$$\arctan\left(\frac{y}{x}\right) \geq \arctan\left(\frac{y_i}{x_i}\right) \wedge \arctan\left(\frac{y}{x}\right) < \arctan\left(\frac{y_{i+1}}{x_{i+1}}\right)$$

Where, (x, y, z) is the target coordinate, (x_i, y_i, z_i) is the coordinate of the i -th RSU, $(x_{i+1}, y_{i+1}, z_{i+1})$ is the coordinates of the $i + 1$ th RSU. If this equation is satisfied, it indicates that the target has just passed through the i -th RSU at the initial position. Then use $RSU_{i-1}, RSU_i, RSU_{i+1}, RSU_{i+2}$ as a sub-cluster of RSUs, where RSU_i is the Master RSU.

4.5 Positioning system

4.5.1 System with DEM and Laser range (RSU-LR-DEM)

The structure and process of the positioning system based on RSU under weak GNSS signals are shown in the Fig 4-5.

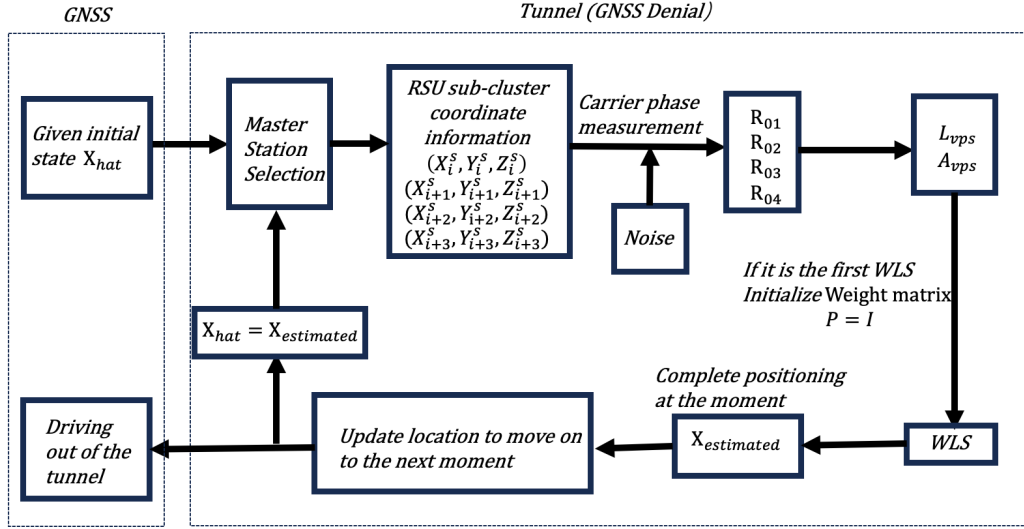


Fig 4-5 positioning system based on RSU

The System with DEM and Laser range is built on the basis of System based on RSU, with the basic framework being consistent. However, in the process of establishing total observation equation, laser side radar and DEM are added, and the difference is shown in Fig 4-6.

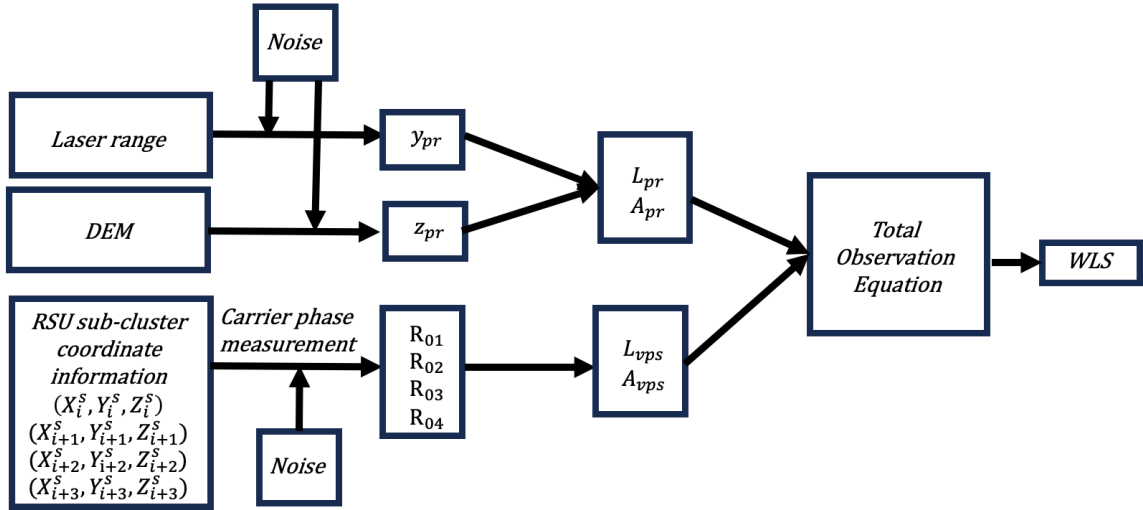


Fig 4-6 positioning system with DEM and Laser range

4.5.2 System with INS and Kalman Filter (INS-KF)

The System with INS and Kalman Filter is built on the basis of System with DEM and Laser range, with the basic framework being consistent. The difference lies in that after WLS regression, the final $X_{estimated}$ is obtained by adding an IMU module and performing Kalman filtering, as shown in Fig 4-7.

The addition of the IMU module enables the system to calculate dynamic information of targets such as velocity and acceleration, achieving real-time dynamic positioning. The addition of Kalman filtering can make the estimated values of position, velocity, and acceleration more stable and reliable.

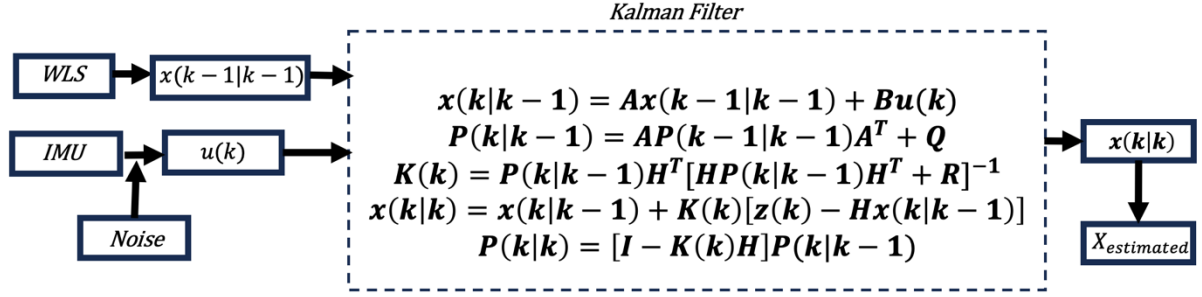


Fig 4-7 positioning system with INS and Kalman Filter

Where, A is the State transition matrix, B is the Control input matrix, x is the target position and motion information(include velocity and acceleration), u is the acceleration measured by the IMU sensor in three directions, H is the conversion matrix, Q is the process noise covariance, R is the measurement noise covariance, K is the Kalman gain, and P is the covariance.

According to the kinematic formula:

$$x = vt + \frac{1}{2}at^2, v = v_0 + at \quad (4.5.1)$$

State transition matrix A , Control input matrix B are:

$$A = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, B = \begin{bmatrix} \frac{1}{2}\Delta t^2 & 0 & 0 \\ 0 & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 0 & \frac{1}{2}\Delta t^2 \\ \Delta t & 0 & 0 \\ 0 & \Delta t & 0 \\ 0 & 0 & \Delta t \end{bmatrix}$$

The independent system error between the speed and displacement is very small, so set $Q = I$. Initialize $P = I$. R is related to the error function set, so it is set as:

$$R = 10 * \begin{bmatrix} \delta_1 & 0 & 0 \\ 0 & \delta_2 & 0 \\ 0 & 0 & \delta_3 \end{bmatrix}$$

Where δ_1 is maximum error of carrier phase measurement φ , δ_2 is maximum error of y_{pr} , δ_3 is maximum error of z_{pr} .

5 Evaluation

5.1 Evaluating indicator GDOP

Assuming that the errors in coordinate positioning are independent of each other in various directions, the covariance matrix of positioning errors can be expressed as:

$$P_{dx} = E \left(\begin{bmatrix} dx \\ dy \\ dz \end{bmatrix} \cdot \begin{bmatrix} dx \\ dy \\ dz \end{bmatrix}^T \right) \quad (5.1.1)$$

And the positioning accuracy GDOP is:

$$GDOP = \sqrt{tr[P_{dx}]} = \sqrt{dx^2 + dy^2 + dz^2} \quad (5.1.2)$$

5.2 Position and Velocity estimation

This chapter will evaluate the performance of the final overall system proposed in this paper, the INS-KF synchronous positioning system, based on the estimated fitting degree in 3 direction, GDOP, velocity fitting degree, and algorithm running time consumption. In Mode_1 and Mode_2, true value of position and the estimated position of the RSU-LR-DEM and INS-KF system in X directions are shown in the Fig 5-1~2 and estimation in X and Y direction of Mode_3 is shown in Fig 5-3:

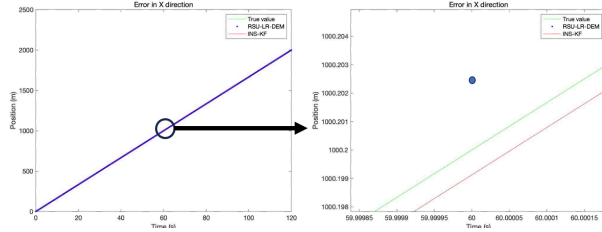


Fig 5-1 estimation in X direction of Mode_1

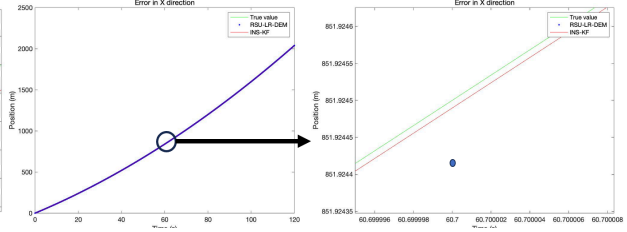


Fig 5-2 estimation in X direction of Mode_2

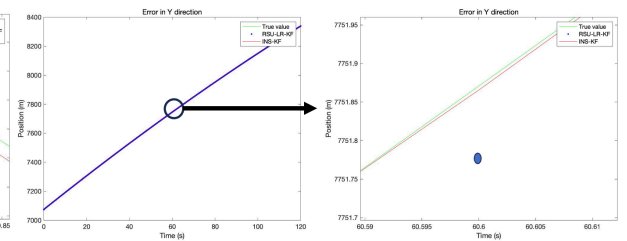
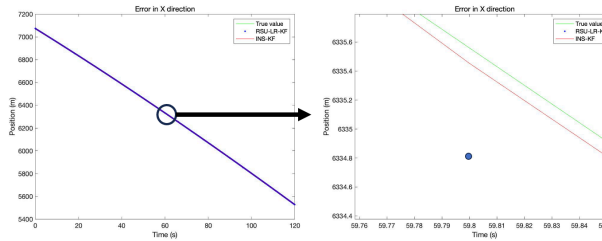


Fig 5-3 estimation in X and Y direction of Mode_3

In Mode_1 and Mode_2, true value of position and the estimated position of the RSU-LR-DEM and INS-KF system in Y directions are shown in the Fig 5-4:

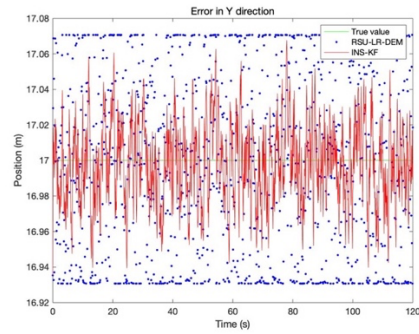
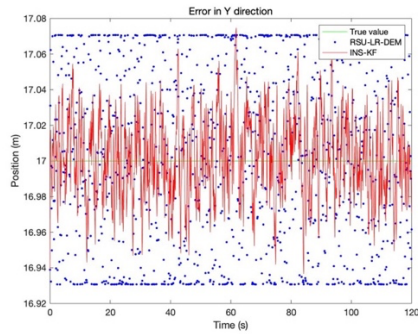


Fig 5-4 estimation in Y direction of Mode_1 and Mode_2

In three simulation experimental modes, true value of position and the estimated position of the RSU-LR-DEM and INS-KF system in Z directions are shown in the Fig 5-5:

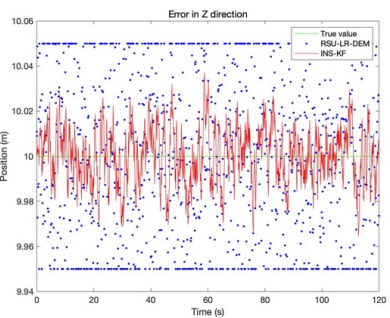
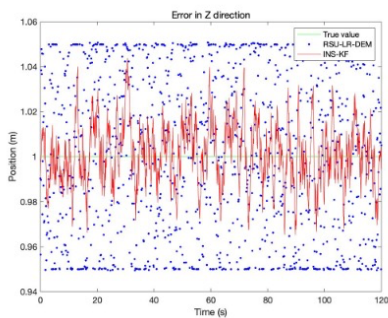
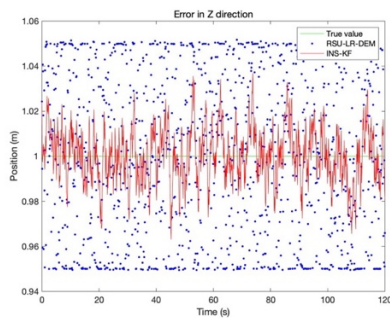


Fig 5-5 estimation in Z direction of Mode_1, Mode_2 and Mode_3

In 3 simulation experimental modes, the estimated velocity v_x , v_y , v_z of the are shown in Fig 5-6~7:

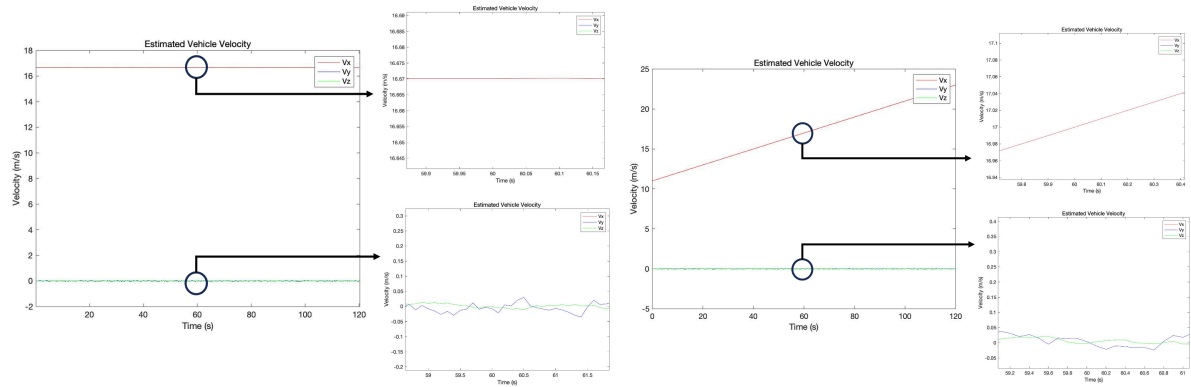


Fig 5-6 estimated velocity in Mode_1,Mode_2

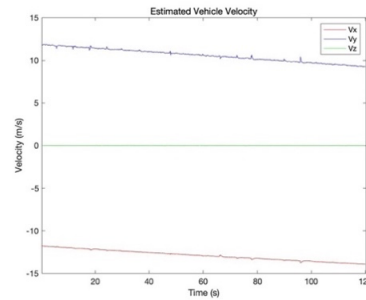


Fig 5-7 estimated velocity in Mode_3

5.3 Estimation Error and Time consumption

According to the calculation method of GDOP, the overall GDOP of the positioning system is calculated, which can reflect the accuracy of positioning. The lower the GDOP, the better the positioning performance and the higher the accuracy. The overall GDOP over time under 3 simulation experimental modes is shown in Fig 5-8~10:

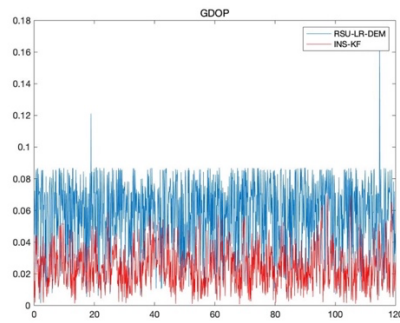


Fig 5-8 GDOP in Mode_1

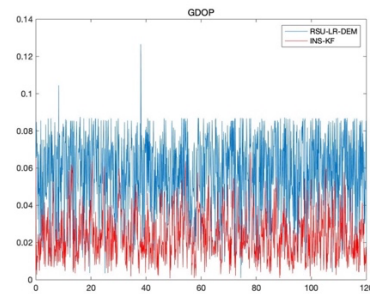


Fig 5-9 GDOP in Mode_2

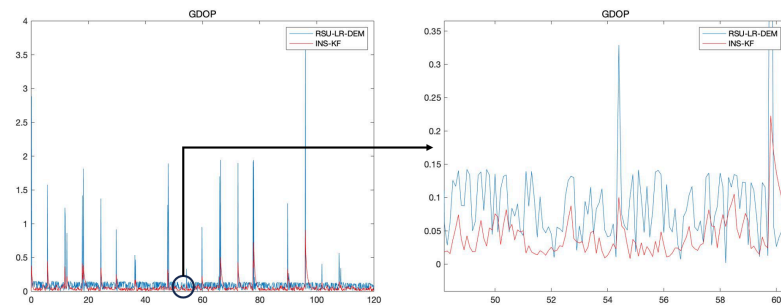


Fig 5-10 GDOP in Mode_3

The average GDOP and average time consumption under 3 modes are shown in the Fig 5-11~12:

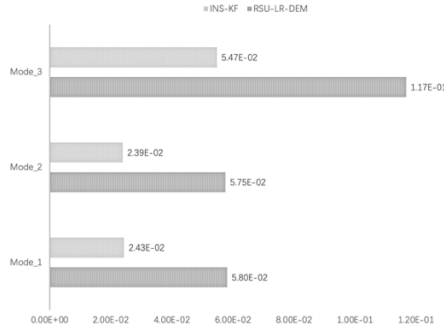


Fig 5-11 average GDOP

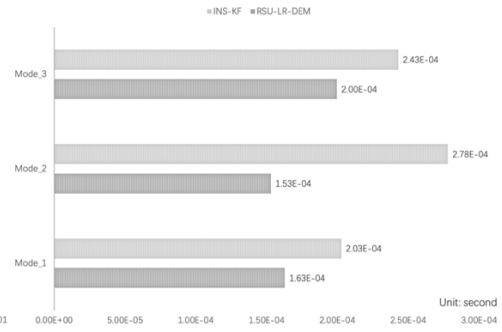


Fig 5-12 average Time Consumption

6 Conclusion

6.1 System performance evaluation

- 1) In terms of positioning performance, the system can accurately estimate the real-time position information of the target, achieving real-time positioning function in weak GNSS scenarios, and accurately reflecting the target's motion trajectory and mode. The INS-KF system can also estimate the speed of the target vehicle with excellent fitting. However, there are still problems where the error suddenly increases during the switching of the main base station, leading to a sudden decrease in positioning accuracy, and even causing non convergence of regression estimation and unstable continuous positioning.
- 2) In terms of accuracy, in three simulation experimental modes, the positioning accuracy of the RSU-LR-DEM system is about 5cm~10cm, and the positioning accuracy of the INS-KF system is about 2cm~5cm. The system performs exceptionally well in terms of positioning accuracy, achieving centimetre level positioning accuracy and meeting industry lane level (meter level) positioning requirements[18].
- 3) In terms of calculation time consumption, in three simulation experimental modes, the average time consumption of the estimation algorithm for RSU-LR-DEM system is about 0.15~0.2 milliseconds, and the average time consumption of the estimation algorithm for INS-KF system is about 0.2~0.3 milliseconds, achieving high-precision and low latency positioning function.

6.2 Further problem

- 1) Multipath effect

The generation of multipath is caused by the reflection of signals. In the application of pseudo satellites, especially indoors, there are more reflections (such as on walls, tables, chairs, etc.), and the pseudo satellites themselves also reflect signals, which makes multipath effect a major technical issue. Generally, in a stationary environment, the multipath of a PS remains unchanged and can be calibrated in advance, such as modelling the error through tracking algorithms in receiver software and determining the deviation value of the error before calibration[19]. However, in the case of movement, the multipath changes are large and random, making the problem difficult to handle. The current solutions to multipath effects include improving the antenna; Data filtering and adaptive processing; Space time combination; Select the optimal constellation layout and receiver position; Improve hardware design including receiver, receiver antenna, and PS signal transmitter antenna.

- 2) Far-Near Problem

Compared with navigation satellites, pseudo satellites are very close to users and have a larger range of relative distance variation. When the distance is close, the power of the pseudo satellite signal is more likely to overwhelm the satellite signal, making it unable to track the satellite signal; When the distance is too far, the signal power is too weak, and the receiver cannot track the pseudo satellite signal. This is the so-called far near problem. The main methods to solve the far near problem of pseudo satellites include trajectory constraints; Design antenna pattern; Using discrete antennas; Frequency modulation; Select new spreading codes; Out of band emission; Frequency offset; Pulse emission, etc.

3) Tropospheric delay

The atmospheric delay of GPS signals includes tropospheric delay and ionospheric delay, while pseudo satellites are generally located near the Earth's surface due to their low altitude, so we only consider the impact of tropospheric delay. GPS satellite signals are usually corrected or eliminated using Saastamoinen, Hopfield, or Black models, but these models depend on satellite elevation, and pseudo satellite elevation is generally too low. Continuing to use these models in pseudo satellite systems can bring significant errors. The existing models for solving the tropospheric delay of PS include Hein's function model that describes atmospheric refractive index as a meteorological parameter; The adaptive tropospheric delay estimation algorithm proposed by Bartrop; The RTCA model of LASS for aircraft precision approach and landing proposed by RTCA and its modified version by Biberger et al; The Bouska model proposed by Bouska et al[20].

In addition, pseudo satellites also have positional errors and nonlinear effects; Hardware design of pseudo satellites and production of supporting user receivers; Selection of PS positions and quantities; Key technical issues such as allocation of new bands and other signal interference.

6.3 Future improvement

- 1) Adding GRU method for data fusion to further improve positioning accuracy[21]: Machine learning methods such as GRU can be used for positioning. Based on the position and motion information of the target, a model for precise positioning and motion state estimation prediction can be trained. In unconventional weak GNSS scenarios (such as irregularly shaped roads, indoor environments, etc.), machine learning can be used for positioning. Weighted or voting methods can also be used to fuse data with existing systems, further improving positioning accuracy and increasing positioning generalization ability, expanding system usage scenarios.
- 2) Further research on vertical positioning performance: Add sensors in the vertical direction, such as barometers and vertical Hall sensors, to further study the vertical positioning performance and meet the positioning needs in uphill, mountainous, and sea scenes.
- 3) Increase the robustness of the algorithm: Consider using non-linear regression estimation methods and modifying the judgment values of fitting regression stability to reduce computational time consumption, increase computational efficiency, and improve the robustness of the algorithm.

7 References

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8 Appendices

A. Record of Meetings Form